Research Design Memo: EU AI Regulation Timing and U.S. Federal AI/Cloud Procurement

Hanning Zhou

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Attached materials.

- **Code:** `code/analysis_final_memo_authentic.py` — data ingestion, cleaning, panel construction, and OLS with firm fixed effects and HC1 standard errors.
- **Prompts:** `prompts/final_memo_prompts.md` — optional prompts for a future GRI v2 workflow, memo-alignment notes, and a data-handling note.

Executive Summary

This memo asks whether major EU AI Act milestones coincide with changes in U.S. federal procurement flows to a small group of AI and cloud vendors. The empirical strategy combines a quarter-level timing indicator tied to the European Parliament's June 2023 vote, transaction-level USASpending extracts covering prime awards and sub-awards, and a simple text measure, the geopolitical rhetoric index (GRI), derived from contract descriptions. After aggregating 2,579 transactions into a 70-row firm–quarter panel spanning 2019Q1–2025Q3, the data show a noticeable increase in average quarterly obligations after 2023Q2, especially for AWS. A regression with firm fixed effects and HC1 standard errors also produces a positive, statistically significant coefficient on the post-milestone indicator, while the lagged keyword-based GRI contributes little explanatory power. These results are best interpreted as descriptive rather than causal. The same period also includes major shifts in U.S. defense cloud spending and procurement modernization, so the memo's main contribution is to map the pattern clearly and show how it can be studied more rigorously in a larger design.

1 Research Question

Core question:

> How does **transatlantic regulatory divergence around AI**, specifically the **timing of EU AI Act legislative milestones**, relate to **realized U.S. federal procurement flows** (prime and sub-award obligations) to **selected AI and cloud vendors**, conditional on observable **mission-oriented framing** in contract narratives?

The political signal in the design is the timing of the European Parliament's **June 2023 negotiating-position vote**, which marked a visible step toward the eventual AI Act framework. The economic outcome is quarterly federal obligations to named vendors in USASpending transaction records, measured in U.S. dollars at the firm–quarter level.

This framing keeps the question close to market-relevant quantities. If the "Brussels Effect" pushes vendors toward common compliance standards, product redesign, or geographic segmentation, investors would want to know whether those pressures coincide with weaker public-sector demand, offsetting demand, or simply a reallocation across business lines. U.S. national-security procurement is therefore useful not only as an outcome in its own right but also as a possible hedge channel against regulatory pressure coming from Europe.

2 Existing Approaches

Several literatures and industry practices bear on the link between **foreign AI governance** and **firm-level revenue exposure**, though few combine EU milestones with granular U.S. obligating actions.

2.1 "Brussels Effect" and extraterritorial regulation. Legal and IR scholarship (e.g., Bradford's framing of EU regulatory export) emphasizes how EU rules reshape global product standards when firms unify compliance to access the Single Market. Empirical work often studies firm adoption of privacy or product rules or equity-market reactions to headline regulatory events. For AI, analysts track draft text, trilogue outcomes, and final risk-tier lists; event-study abnormal returns around key votes are common in sell-side and policy notes.

2.2 Event studies and difference-in-differences. Applied work in law, economics, and CS policy uses staggered adoption, EU vs. non-EU comparisons, or sector exposure scores from 10-K and supply-chain data. Outcomes include stock returns, patenting, job postings, or Model lists compliance. Some designs exploit national transposition timing or court decisions.

2.3 Vendor-level public procurement analytics. Researchers use USASpending, FPDS, or SAM data to trace award families, IDIQs, and agency programs (e.g., cloud GWACs). Practitioners build vendor revenue bridges from transaction-level obligations to investor filings where possible. Text fields are mined for NAICS, PSC, program acronyms, and mission language.

2.4 Text-as-data measures. Political science and computational social science extract frames from speech, media, or corporate disclosure. A simple keyword index is a baseline; dictionaries, topic models, and increasingly LLM structured extraction reduce polysemy but raise reproducibility and PII concerns with raw government text.

2.5 Sell-side "policy beta" and ESG overlays. Equity research teams sometimes construct explicit exposure scores to data localization, AI ethics, or sanctions, then correlate them with returns or forward estimates. These overlays are valuable for portfolio-facing summaries but often aggregate at sector level or rely on subjective disclosure coding unless paired with transaction facts such as obligations.

Taken together, these literatures point toward a hybrid design: use EU legislative timing to define the political backdrop, use procurement transactions to observe realized commercial exposure, and use text analysis to recover variation in how awards are framed. None of those pieces is sufficient on its own, but combined they make it possible to study a vendor-level channel that is usually discussed only in broad sector terms.

3 Limits of Existing Approaches

3.1 Identification. EU AI Act milestones are not the only exogenous shocks to U.S. procurement; the same period saw JWCC rollout, Ukraine conflict–driven modernization, CHIPS-era industry policy, and Fed tightening. A calendar dummy after a Parliament vote picks up joint trends, not marginal effects of Brussels on U.S. obligation streams.

3.2 Cross-border mapping. The Brussels Effect predicts global product harmonization, not necessarily lower U.S. public spending. Firms may dual-track offerings: EU-compliant SKUs vs. U.S.-only missions. Aggregate equity reactions may miss segment-level procurement.

3.3 Data granularity vs. noise. USASpending transaction obligations vary with modification timing, IDIQ contracting mechanisms, and reporting errors. Single-line outliers can dominate small vendor panels. Sub-award descriptions may be thin or mis-keyed.

3.4 Text measurement. Keyword "security" hits cybersecurity patches as often as national security. Without context-aware scoring, GRI v1 is high-noise.

3.5 Coverage. Focusing on three firms is useful for a tightly documented pilot analysis, but it does not support representative inference for the broader population of AI-adjacent federal suppliers.

3.6 Temporal misalignment. Contract action dates reflect ordering activity, not necessarily EU board decisions or product redesign completion. A vendor may obligate under an IDIQ option years after an EU vote; conversely, regulatory anticipation may shift behavior before the milestone encoded here.

3.7 Firm FE vs. firm-specific trends. The panel covers only a modest number of quarters for each firm, and the number of firms is very small. In that setting, adding firm-specific linear time trends could absorb variation that the current specification attributes to the post-milestone indicator D_t. At the same time, firm fixed effects alone do not absorb common macro shocks that move with the EU timeline.

4 Research Design: "PR–Procurement Tracker"

In response to those limits, the research design emphasizes transparency and a sequence of choices that can be checked directly in the data. Rather than introducing a complicated identification strategy too early, I begin with a simple panel that makes the timing assumptions visible and keeps the link between raw transactions and reported results easy to trace. The workflow proceeds in seven steps:

1. Read firm-specific USASpending CSV extracts under `data`, covering prime contracts, prime assistance, and sub-awards, and harmonize the relevant identifier, date, amount, and description fields across file types.

2. Restrict the sample to four validated UEIs that map to three firms: one UEI for Palantir, two UEIs merged into C3.ai, and one UEI for AWS.

3. Convert action dates to calendar quarters, drop records with missing quarter information, and trim any single-line obligation above 10^{10} USD as an obvious reporting artifact.

4. Construct the geopolitical rhetoric index, or GRI, as the share of transactions in each firm–quarter whose description contains at least one keyword from a fixed list (*defense, security, intelligence, warfighting, surveillance*).

5. Lag that GRI measure by one quarter within each firm so that the text-based signal precedes the procurement outcome it is used to explain.

6. Define a post-milestone indicator, written as D_t, that equals 1 from 2023Q3 onward and 0 beforehand. Here, t denotes the calendar quarter, so D_t simply marks whether a given quarter falls after the June 2023 parliamentary milestone.

7. Estimate a firm–quarter OLS regression with firm fixed effects implemented as indicator variables; under `pandas drop_first` ordering, AWS is the omitted reference category.

Model setup and notation. Before presenting the equation, it helps to define the notation in plain language. The unit of analysis is a firm–quarter observation. The subscript i identifies the firm and the subscript t identifies the calendar quarter. The outcome variable, Y_{it}, is total federal obligations to firm i in quarter t, measured in millions of U.S. dollars. The term α_i captures firm fixed effects, or time-invariant differences across firms. The variable $\mathrm{GRI}_{i,t-1}$ is the one-quarter lag of the geopolitical rhetoric index for firm i. The indicator D_t equals 1 in quarters from 2023Q3 forward and 0 before that point. Finally, ε_{it} is the residual term.

With those definitions in place, the estimating equation is

$$Y_{it} = \beta_0 + \alpha_i + \beta_1 \,\mathrm{GRI}_{i,t-1} + \beta_2 D_t + \varepsilon_{it} \,. \tag{1}$$

Definition of the geopolitical rhetoric index. For each firm i and quarter t, let m_{it} denote the number of transactions with at least one keyword match and let n_{it} denote the total number of transactions. The geopolitical rhetoric index is then defined as

$$\mathrm{GRI}_{i,t} = \frac{m_{it}}{n_{it}} \,. \tag{2}$$

In the regression, I use the lagged value $\mathrm{GRI}_{i,t-1}$ so that the text measure is observed before the quarter's outcome.

This design is best understood as a clear empirical baseline. It standardizes heterogeneous procurement files, produces a panel that can be inspected directly, and makes the modeling assumptions explicit. It does not, by itself, deliver a final causal estimate, but it does move the discussion beyond general claims about regulatory risk and toward a replicable measurement strategy.

Operational workflow. The script `code/analysis_final_memo_authentic.py` recursively ingests CSV files, routes each file to a loader based on whether it is a prime-contract, prime-assistance, or sub-award extract, standardizes column names to a common schema, coerces obligations to numeric values, fills missing descriptions with empty strings, filters to the validated UEI list, and drops rows without a parsable action quarter. It then aggregates transactions to the firm–quarter level, counts total transactions and keyword matches, computes the contemporaneous GRI share and its within-firm lag, rescales obligations into millions of dollars, and estimates OLS with HC1 heteroskedasticity-robust covariance. Because the panel is small and visibly unbalanced across firms, the script also reports pre/post means so that the regression can be read alongside the raw descriptive shift.

Why this design is useful. The purpose is not to propose a new theory of the Brussels Effect. The value of the design is that it links an external regulatory timeline to observable U.S. procurement outcomes at vendor level and makes it easy to replace the simple GRI with a more context-aware version later.

5 Preliminary Data Analysis and Results

Sample and span. The cleaned transaction file contains 2,579 observations from 2019Q1 through 2025Q3. After aggregation, those records produce 70 observed firm–quarter cells:

27 for AWS, 27 for Palantir, and 16 for C3.ai. Vendor identities are defined by validated UEIs: Palantir (`FSY4LVSBGWB7`), C3.ai (`YT9GYGR24NW9` and `YYRJSL45NMQ7` combined), and AWS (`NQEWN6C1LSU5`). The panel is therefore informative about within-firm time variation, but noticeably sparser for C3.ai than for the other two firms.

Descriptive shift (mean quarterly obligations, USD millions). The first pattern in the data is a strong pre/post level shift around 2023Q3. After removing the single implausible Palantir sub-award artifact, mean quarterly obligations rise for all three firms:

Firm	Pre-shock mean ($M/qtr)	Post-shock mean ($M/qtr)
AWS	∼22.7	∼125.3
Palantir	∼45.6	∼68.5
C3.ai	∼1.5	∼3.3

These means are calculated over the quarters observed for each firm. Because the panel is unbalanced, they should be read as descriptive summaries of realized activity rather than as directly comparable exposure rates. In level terms, AWS accounts for the largest post-shock expansion: total obligations rise from roughly $0.41B before 2023Q3 to about $1.13B afterward. Palantir remains large in both periods, with about $0.82B pre-shock and $0.62B post-shock, but its post-shock mean still increases because the later period covers fewer quarters with larger average actions. C3.ai remains much smaller in absolute dollars, moving from roughly $12M pre-shock to $26M post-shock.

Qualitative anchors (major vehicles).

- **AWS:** Joint Warfighting Cloud Capability (JWCC) language on DISA/WHS vehicles (e.g., family `HC105023D0005`).
- **Palantir:** Army Capability Drop 1 delivery orders under Army `W56KGY18D0003` (2025 obligations).
- **C3.ai:** Air Force SBIR Phase III database support (`FA460019CA025`) with modifications referencing SBIR Phase III.

These qualitative anchors help explain why the post-2023Q2 increase is not naturally attributable to EU regulation alone. Much of the visible growth in the procurement data is tied to mission programs, cloud vehicles, and defense modernization pathways that have their own U.S. budget and operational logic.

Regression (HC1 robust SE). Once the one-quarter lag of GRI is imposed, the estimating sample falls from 70 observed firm–quarter rows to 67. AWS is the reference firm because its indicator is omitted from the regression design matrix.

Variable	Coef.	Robust SE	z	p
Constant	42.319	8.984	4.71	0.000
$\text{GRI}_{i,t-1}$	11.838	26.040	0.46	0.649
D_t	**47.187**	13.709	**3.44**	**0.001**
C3.ai vs. AWS	−66.611	15.254	−4.37	0.000
Palantir vs. AWS	−5.667	13.806	−0.41	0.681

$R^2 = 0.358$; Adj. $R^2 = 0.317$.

Interpretation. Three points matter most. First, the coefficient on D_t is positive and economically large: conditional on firm fixed effects and lagged GRI, quarters after 2023Q2 are associated with roughly \$47M more in obligations per firm–quarter. That estimate is statistically precise in this sample, but it should still be read cautiously. The post-2023Q2 period also captures expanding U.S. defense cloud procurement, especially in AWS-linked vehicles, as well as broader procurement normalization after earlier disruptions. Second, the coefficient on lagged GRI is positive but highly imprecise. In practical terms, the keyword index explains little additional variation once firm baselines and the common post-milestone period are included in the model. That fits the underlying data: keyword hit rates are low for AWS (about 3.6% of transactions overall), modest for Palantir (about 6.9%), and higher for C3.ai (20%) but based on a very small number of transactions. Third, the firm coefficients mainly capture level differences rather than deep structural contrasts. C3.ai sits far below AWS in obligation levels, while Palantir is statistically similar to AWS once the common post-period shift is taken into account. In short, the regression is most useful as a compact summary of patterns already visible in the descriptive evidence, not as a decisive test of a Brussels-driven causal mechanism.

6 Discussion

The evidence in this memo points to two channels that should be kept conceptually separate. On the European side, stricter rules on biometric surveillance and other high-risk AI uses can raise compliance costs, encourage product segmentation, and change where vendors choose to market or deploy particular systems. On the U.S. side, the procurement narratives in this sample point toward a different dynamic: warfighting cloud, Army intelligence modernization, and SBIR Phase III pathways have their own budgetary and operational momentum, much of it independent of Brussels.

The main takeaway is therefore not that EU regulatory divergence mechanically reduces U.S. procurement flows. A more defensible reading is that regulatory divergence increases strategic complexity. It can reshape product strategy and compliance burdens while U.S. mission demand continues to rise through separate channels. That is why a scenario-based interpretation is more credible here than a single reduced-form causal claim.

7 Next Steps

7.1 Causal front door: Merge firm-level EU revenue shares with fine-grained AI Act instruments; estimate triple-differences (high-EU exposure × high-surveillance product × post-milestone) with firm and calendar FE, or Sun–Abraham staggered designs where appropriate.

7.2 Parallel trends diagnostics: Plot firm-specific cumulative obligations pre 2023Q3 against placebo shock dates; run TWFE bias diagnostics when D_t is essentially common time.

7.3 Expand the cross-section: Additional cloud/AI primes with consistent UEI parenting rules (`recipient_parent_uei` vs. immediate recipient).

7.4 GRI v2 with LLMs: Freeze model ID, temperature, and prompt hash; label unique descriptions once; human audit on a stratified sample (see `prompts/final_memo_prompts.md`).

7.5 Outliers and linkage: Award-family ceilings from parent PIID distributions; join to FPDS for pricing and award-type filters.

7.6 Forecasting track: Lag-GRI, agency dummies, and macro controls to forecast next-quarter obligations out-of-sample; report RMSE by firm.

8 Reproducibility

All numerical results in Sec. 5 were produced by running `python code/analysis_final_memo_authentic.py` from the project root on the bundled `data` extracts. In the current version of the pipeline, text processing is limited to deterministic keyword matching using a fixed regex compiled from the GRI dictionary; no LLM is applied to raw federal text. Optional prompts for a future GRI v2 workflow are stored in `prompts/final_memo_prompts.md`.

End of research design memo.